FinPersona-Bench: Synthetic Environment v2, Research-Grounded Reimplementation Plan

Purpose: a justifiable specification for rebuilding envs/synthetic_market.py, the harness and the evaluation layer before implementation begins. Three research threads were run (allocation targets and persona realism; time-series realism; design precedents and validation), each with primary sources read in full or fetched from the web, plus a diagnostic run of the current generator (20 seeds x 200 days). Version 3 (22 August 2026) integrates an author review of the first draft (items A1 to F29, Section 12), an independent fact-check of every cited number against primary sources, and a coherence review (67 findings); the fact-check and coherence reports are in docs/env_research/ beside the three research reports. Timelines are deliberately omitted. Conventions: T = 200 trading days for the main grid (T = 800 for the phase-free control); C denotes the CASH share of the portfolio everywhere; a 260-day burn-in precedes day 1 so that trailing-four-quarter earnings, SMA50 and the GARCH state are warm at t = 1.

1. Diagnosis of the current generator and harness (what is actually wrong)

From reading envs/synthetic_market.py, simulation/portfolio_tracker.py and simulation/runner.py in full, the internal audit (docs/SYNTHETIC_MARKET_AUDIT.html, 14 Aug 2026), a diagnostic run re-executed during the coherence check, and the author review. Every item below is verifiable from the repository; all line numbers and code claims were re-checked by the fact-checker.

#	Defect	Evidence	Consequence
1	Ground truth leaks exactly. EPS = V/15 every day, so reported P/E = 15 P/V and dividend yield = 0.4 EPS/P both invert to V.	Code lines 230 to 246; audit F-1: max reconstruction error 0.0000; sign(P minus V) = sign(PE minus 15) on 100 percent of steps.	'Hidden fundamental value' is false as implemented; RG partly measures P/E-prior alignment.
2	Phases are fixed day indices (40/30/30 percent of T) for every seed; volatility, volume, sentiment and implied volatility follow the same clock.	Lines 58 to 61, 104 to 109, 145 to 150, 194 to 217; IV identical across seeds in crash and bull.	Elapsed time and market phase are perfectly confounded; IV and volume are phase clocks an agent can read.
3	No volatility clustering and no fat tails. The 'GARCH-like' recursion is phi = 0.7 phi + 0.3 |N(0,1)| scaling an additive 0.5 dollar noise; it never feeds back from returns.	Diagnostic, flat: excess kurtosis 0.06, ACF of |r| at lag 1 = 0.01, corr(volume, |r|) = minus 0.02. Paper claims GARCH (Bollerslev 1986).	None of Cont's stylized facts hold; Appendix A misdescribes the code.
4	Additive dollar noise around a level creates a negative return autocorrelation (bid-ask-bounce signature) and a free mean-reversion signal.	Diagnostic, crash: ACF(1) of returns = minus 0.45 for all three deltas.	Detectable artefact; agents can infer mispricing from return statistics alone.
5	The crash severity parameter delta does almost nothing.	Max drawdown minus 47.5 / minus 47.2 / minus 47.0 percent for delta 0.85 / 0.92 / 0.95; the deterministic minus 12 dollar decline plus the panic random walk in V dominates.	Appendix J's 'sensitivity analysis' manipulates almost nothing.
6	Bull-trap mispricing is a	Lines 86 to 100; daily sigma 1.3 percent in	Bubble has no theoretical

	cumulative dollar random walk, so P/V reaches 2.2 with lower volatility than the flat market and volume is negatively correlated with |r|.	bull vs 2.1 flat; corr(volume, |r|) = minus 0.21.	provenance and inverts the volume-volatility relation.
7	Hard clips ('structural invariants') at P >= 1, V >= 10, P <= 0.98 V, blow-off >= 1.05 x mania entry.	Lines 83, 89, 100, 123, 137 to 143; audit F-10.	Kinks and deterministic signals; no acceptance rate is reported.
8	Observation-space documentation mismatches. Table 2 lists MACD and dividend yield that never reach the prompt; Table 2 omits the portfolio state (cash, holdings) that the prompt does include; the key 'SMA60' is the 50-day average.	Lines 250 to 261; static_agent.py lines 122 to 124; audit D-6/D-7.	Paper, prompt and code disagree.
9	In the flat scenario 96.9 percent of steps have |P minus V| / V below 1 percent.	Audit F-2.	The 'objectively correct action' is inside generator noise for almost the whole flat horizon.
10	No transaction costs, fractional shares, same-day execution at the observed price; none declared.	portfolio_tracker.py; survey MR-3/MR-4.	Churn is free; 'trading without signal' has no economic cost.
11	No baselines, stylized-facts tests, leakage probe or reliability accounting in the environment or the paper.	Audit F-3 (computed by the audit, not the paper): buy-one-share-then-HOLD RG 99.7 in the bull trap; always-HOLD 79.8; random 66.9; both reported arms (85.9 / 78.4) sit inside the 66.9 to 99.7 envelope.	Reported metric levels are uninterpretable.
12 (review A1)	Every run starts at 100 percent cash. ISFJ (target 1.0) starts exactly at target, so its MAS is 0 at t = 0 and can only rise; ENTJ (target 0.2) starts at the maximum possible deviation (0.8) and can only fall.	portfolio_tracker.py: initial_cash = 10000.0, holdings_qty = 0.0. Day-1 cash 0.88 / 0.92 / 0.92 across personas (Cycle 2); ENTJ static cash 0.39 in Q1 falling to 0.24 by Q4.	Combined with the portfolio ratchet (buys convert cash to equity, little converts back) the initial condition alone produces ISFJ 'drift up', ENTJ 'drift down', and makes any cash-raising intervention help ISFJ and hurt ENTJ by construction. A cleaner alternative explanation for the bidirectional headline than anything in Cycle 1, and an environment defect.
13 (review A2)	BUY and SELL are asymmetric operators: BUY spends cash x quantity, SELL sells holdings x quantity. From a 100 percent cash start SELL is a no-op on day 1, and repeated fractional buys approach full investment geometrically without overshooting.	portfolio_tracker.py lines 40 to 68.	Interacts directly with 12 and the ratchet; the action semantics are undeclared in the paper.
14 (review C12)	The horizon is not disclosed. The agent sees 'Day-N' but is never told T.	synthetic_market.py get_observation 'date': f'Day-{day}'; no T in the prompt templates.	Terminal-horizon effects change optimal allocation; undeclared factor.
15 (review C16)	The ISFJ mandate instructs an action outside the action space: 'Buy insurance (puts/hedges)' against {BUY, SELL, HOLD} on one asset.	agent/personas/mbti_profiles.json; Appendix B.2.	Unexecutable instruction inside the mandate being scored.

2. The design decision

Four options were compared against the precedents (Section 10). Recommendation: the hybrid (C), built N-asset-capable from the start so that the multi-asset extension (D) is a configuration rather than a rewrite. Two facts must be stated up front because they bound what the hybrid can honestly claim: (i) the price path is exogenous to the agent (zero market impact; the Franke-Westerhoff 'market maker' is a generative time-series model, not a counterparty), and (ii) the fundamentalist-chartist process shapes the DYNAMICS of mispricing (persistence, volatility clustering, overshoot) in every phase, but the DIRECTION and TARGET of event phases (a panic discount, a bubble run-up) are imposed by scripted drifts. The environment is 'FW-shaped, regime-scripted', not 'FW-generated'.

Option	For	Against	Verdict
A. Scripted but stochastic (minimal patch)	Cheap; seed-reproducible regimes; V independent of agent actions.	Mispricing dynamics have no provenance; answers nhWU's 'unjustified constants' only partly.	Fallback. Must still include real GARCH, randomised onsets, leak fix, baselines, stylized-facts table.
B. Endogenous ABM (Franke-Westerhoff market maker, Lux-Marchesi, Chiarella-Iori order book as in Hashimoto/PAMS, ABIDES)	Bubbles and crashes emerge from calibrated demand; stylized facts come with published parameter sets.	Regimes are stochastic: a seed may not produce a bubble in 200 days, so guaranteeing a scenario needs seed selection; the LLM's own orders would have to move price, breaking V-independence and reproducibility; hard to cross with models x personas x seeds.	Not for this cycle.
C. Hybrid (recommended): exogenous fundamental V (GBM); mispricing x = log P minus log V whose calm-phase dynamics follow the Franke-Westerhoff 2012 fundamentalist-chartist recursion and whose event phases add scripted directional drifts; one innovation process with GJR-GARCH-t volatility; scheduled regimes with per-seed onset and duration; scenario validity by referee-side rejection sampling with the full conditioning published.	Seed reproducibility; V and P independent of agent actions; phases as a controlled factor; published provenance for the FUNCTIONAL FORM of the decoupling (the parameters are re-estimated, Section 2.2); stylized facts testable; time separable from phase.	More code than A; the schedule durations are design choices; FW's published parameters are index-level and give multi-year mispricing persistence (Section 2.2); rejection sampling is a form of conditioning and is reported jointly with the probabilistic bubble top (C9).	Do this.
D. Multi-asset extension of C (review E23): 2 to 5 assets, each with its own exogenous V path and mispricing process, with a common factor in the fundamental shocks.	'Single-agent, single-asset' is a named Program-Chair rejection reason; with exogenous per-asset fundamentals you get portfolio effects, cross-asset correlation and a genuine allocation problem without losing ground truth; lets the conservative mandate name a defensive asset (fixes defect 15).	Action space becomes a target weight vector; the cash-share band stays the scored quantity (weights within the risky sleeve are free); resolvability and oracles become per-asset; the defensive asset must be low-volatility RISKY (a risk-free asset would be cash by another name); more tokens per call.	Build the generator vectorised over N assets now; run the main grid at N = 1 for comparability with v1; run a 3-asset scenario as a headline extension. Cost it; do not defer the design.

2.1 Composition of price, and the generator block by block

One decomposition, used everywhere: log P_t = log V_t + x_t, so the price return is r_t = delta log V_t + delta x_t. There is ONE innovation process for x; its conditional scale is a GJR-GARCH(1,1) with Student-t(5) shocks, and the Franke-Westerhoff sigma_f / sigma_c enter only as relative weights of the fundamentalist and chartist groups (normalised to unit mean), not as a second noise source. This resolves the three-source ambiguity flagged by the coherence review.

1. Fundamental V_t: log V_t = log V_t-1 + mu_V + sigma_V z_t, z ~ t(5); mu_V = 0.00025/day (about 6.5 percent a year; long-run price-only equity return 6 to 8 percent), sigma_V = 0.006/day (the fundamental is smooth; the majority of single-stock variance is idiosyncratic mispricing, Campbell, Lettau, Malkiel and Xu 2001). Crash: a log-linear negative drift in the deterioration phase delivering D_V ~ U(10, 30) percent, then flat. Bubble: V keeps growing at mu_V; no artificial plateau (Greenwood, Shleifer and You 2019). Sustained bull: mu_V ~ U(0.0015, 0.0025)/day for the episode. Cite as GBM, not Black-Scholes. Multi-asset: one V path per asset with a common factor in z.
2. Mispricing x_t, calm phases: x_t+1 = x_t + mu (n_f phi (minus x_t) + n_c chi (x_t minus x_t-1)) + sigma_t e_t, with n_f = 1 / (1 + exp(minus beta a_t)), a_t = alpha_0 + alpha_n (n_f minus n_c) + alpha_p x_t^2 (Franke and Westerhoff 2012, DCA-HPM), and sigma_t from the GARCH block. At the published index-level parameters the fundamentalist pull is about mu n_f phi = 0.0012/day, i.e. a mispricing half-life of roughly 580 days, consistent with Poterba and Summers' multi-year transitory component but far longer than a 200-day benchmark; see 2.2 for the decision this forces.
3. Mispricing x_t, event phases: an additive scripted drift d_t is added to the calm recursion. Panic: d_t pulls x toward ln(delta) over the panic length (front-loaded: half the move in the first third), then during stabilisation toward ln(delta_end), delta_end ~ U(delta, 1). Mania: a super-exponential drift g_t with g_t+1 = g_t (1 + kappa), g_0 = 0.002/day, kappa ~ U(0.02, 0.04)/day (so the daily drift, not x itself, compounds), running until the top event or the horizon; the top is a HAZARD, h_t = h_0 exp(b x_t), calibrated so that about 50 percent of seeds top inside the horizon and topped seeds peak at P/V 1.6 to 2.5; after a top, d_t reverses (minus 30 to minus 50 percent in 10 to 30 days). Sustained bull: d_t = 0 (x stays near 0; the rise is in V). Direction and target of events are therefore scripted by construction; FW supplies persistence, clustering through chartist share, and overshoot.
4. Volatility: sigma_t^2 = omega_phase + alpha e_t-1^2 + gamma e_t-1^2 1[e_t-1 < 0] + beta sigma_t-1^2; alpha = 0.05, gamma = 0.08, beta = 0.89 (persistence alpha + gamma/2 + beta = 0.98, shock half-life about 34 days); omega chosen so the unconditional x-innovation is 1.6 percent/day, giving total daily price sigma about sqrt(0.6^2 + 1.6^2) = 1.7 percent (about 27 percent a year, a large-cap single stock); omega multipliers: calm 1, deterioration 1.5, panic 4 (sensitivity 3 to 6), stabilisation 1.5, mania 1.5, blow-off 2, post-top 3, sustained bull 1. Optional rare jump: Poisson 0.004/day, size N(minus 4 percent, 3 percent).
5. Phase taxonomy, used everywhere (observables, multipliers, the L2b class set): calm; deterioration, panic, stabilisation (crash); mania, blow-off, post-top (bull trap); sustained-bull. Macro classes for classifiers: {calm, down-event, up-event, resolution}.
6. Schedule (a scheduled regime sequence, not a fitted Markov chain; durations are design choices with the empirical anchors quoted): setup-first ordering: L1 ~ U(0.25 T, 0.55 T), event length L2 ~ U(15, 70) days (crash) or until the hazard top or horizon (mania, with blow-off defined as the last third of the mania run), remainder = resolution; event-first (mirrored) ordering: a short setup U(5, 20) days, the event, then resolution for the remainder with calm dynamics; plus or minus 5 day onset jitter per observable (so observables disagree slightly about the phase; intended, and included in the L2b design). Anchors: Ang and Timmermann 2012 (monthly S&P high/low-volatility regimes, P = 0.977, Q

= 0.951); Pagan and Sossounov 2003 (bull about 25 months, bear about 15 months); Hamilton 1989; crash legs of 23 to 120 days to trough in 2020 and 2008.

7. Scenario validity (rejection sampling) with criteria stated per scenario: crash: min over the panic phase of x <= minus 0.10 and realised MDD >= 20 percent; bull trap: max x >= +0.30 at some point; sustained bull: x within [minus 0.10, +0.15] throughout and final V/V_0 >= 1.2; flat: no criterion. Report the rejection rate per scenario (target below 5 percent) AND the joint conditioning with the hazard top (share of bull-trap seeds that are topped), and stratify every bull-trap metric by topped and un-topped seeds.
8. Initial allocation (review A1): a controlled factor. Primary design: each persona starts at the centre of its own target band; secondary design: all personas start at a common 0.5 cash share; never 1.0 for everyone again. The swapped-mandate arm starts at the PERSONA's target; the no-mandate trader starts at 0.5. All baselines and floors are computed from the same start allocation as the cell. Consequence acknowledged: under the primary design personas start at different equity exposure in an identical market, so cross-persona comparisons of return-type metrics (drawdown, return, RG) are comparisons of starting exposure; such comparisons are made only within persona (static vs memory, as the paper does) or on the common-start design. Relative MAS (|C_t minus C_0|) is defined for the common-start design only (under start-at-target it coincides with point-MAS).
9. Action semantics (review A2): the action is a target CASH share c* in [0, 1]; the trade is the delta between c* and the current cash share; derived labels for RG scoring: BUY if the equity share rises by more than 1 point, SELL if it falls by more than 1 point, HOLD otherwise; HOLD means no trade (the allocation then drifts with price). Execution at the same-day close P_t on which the observation was rendered (the current convention, stated), with next-open execution as a sensitivity. Cost = 5 bp x |traded value| per trade (Section 8). The v1 interface (fraction-of-side) is kept behind a flag for the A1 x A2 comparison (Section 11.3).
10. Separate RNG streams per component (V, mispricing, GARCH innovations, sentiment, EPS noise, volume, initial allocation, field order) so adding a field never changes an old path; a generator config hash and a prompt hash in every output row (review F28).

2.2 Calibration decision for the mispricing process (review C10; coherence item 8)

Franke and Westerhoff 2012 (JEDC 36:1193 to 1211) estimated the DCA-HPM model on S&P 500 daily returns 1980 to 2007 by the method of simulated moments: phi = 0.12, chi = 1.50, sigma_f = 0.758, sigma_c = 2.087, alpha_0 = minus 0.327, alpha_n = 1.79, alpha_p = 18.43, beta = 1, mu = 0.01, bootstrap J-test p = 32.6 percent on nine moments (lag-1 return ACF, mean |r|, Hill tail index, ACF of |r| at lags 1, 5, 10, 25, 50, 100). Pruna, Polukarov and Jennings 2016 (arXiv 1604.08824) re-estimate with a random-walk fundamental: phi = 0.121, chi = 1.555, sigma_f = 0.592, sigma_c = 1.917, alpha_0 = minus 0.301, alpha_n = 1.990, alpha_p = 22.741. Two consequences: (a) these are INDEX parameters and the target here is single-stock volatility, and (b) at these parameters mispricing reverts with a half-life of hundreds of days. Decision: re-estimate phi, chi, alpha_0, alpha_n, alpha_p by simulated method of moments on a panel of about ten large-cap single stocks (daily, 2000 to 2024), keeping the FW moment set; report the index parameters as a sensitivity. The published J-test is provenance for the functional form, not for the parameters used. The benchmark-relevant persistence is then whatever the single-stock estimate gives; Section 9 item 9 is written for that regime, and the flat market is expected to carry realistic, persistent mispricing of the order of 8 to 20 percent standard deviation (Poterba and Summers 1988: transitory component 15 to 25 percent at index level), which raises flat-market resolvability from 3 percent today to a substantial fraction of steps. That is intended: 'flat' means no scripted regime event, not 'price equals value'.

Franke and Westerhoff 2012 (PDF): www.uni-bamberg.de/fileadmin/uni/fakultaeten/sowi_lehrstuehle/vwl_wirtschaftspolitik/Team/Westerhoff/Publications/2011/JEDC_RF_FW_Fin.pdf | Pruna, Polukarov and Jennings 2016: arxiv.org/abs/1604.08824 | SABCEMM contest: arxiv.org/abs/1812.02726 | Hommes 2006 survey: doi.org/10.1016/S1574-0021(05)02023-X | Poterba and Summers 1988: www.nber.org/papers/w2343

3. Generator spec v2: parameter table

Current value (code) | what the paper claims | empirical anchor with source | v2 value or process | validation statistic. Items marked (approx.) were not verified against a primary source in the fact-check and must be checked before the paper cites them.

Block	Current (code)	Paper claims	Empirical anchor (source)	v2	Validate
Daily vol of value path	sigma 0.02 in flat; 0.01 bull phase 1; 0.002 plateau	GARCH-like	Index about 1 percent/day; single large cap 1.6 to 2.5 percent/day; idiosyncratic share majority (Campbell, Lettau, Malkiel, Xu 2001)	sigma_V 0.6 percent/day; x-innovation 1.6 percent/day; total price sigma about 1.7 percent/day	flat: sigma of log-price returns 1.4 to 2.2 percent; of log V 0.45 to 0.75 percent
Drift	mu 0.0005/day on V	'without significant drift'	Long-run US price-only return 6 to 8 percent a year	mu_V 0.00025/day (calm, crash, bull trap); U(0.0015, 0.0025) in sustained bull	flat: mean 200-day log return about 0.05 plus or minus 0.25
Volatility dynamics	phi = 0.7 phi + 0.3 |N(0,1)|, floor 0.5, on a 0.5 dollar noise	GARCH (Bollerslev 1986)	GARCH(1,1) daily equities: alpha about 0.05 to 0.10, beta about 0.85 to 0.93, persistence 0.95 to 0.99 (Engle 2001: 0.077 / 0.905; Bollerslev 1986; Hansen and Lunde 2005: leverage models beat GARCH(1,1) on IBM); GJR 1993: the asymmetry term is positive and significant (magnitudes in their monthly study are not a daily gamma range; typical daily gamma of order 0.1 is approx.)	GJR-GARCH(1,1)-t(5): alpha 0.05, gamma 0.08, beta 0.89; phase omega multipliers	Ljung-Box on |r| (10 lags) p < 0.01 in >= 80 percent of seeds; ARCH-LM rejects; refit alpha + beta in [0.90, 0.995]
Fat tails	Gaussian; kurtosis about 0	not claimed	excess kurtosis 3 to 20; tail index 2 to 5 (Cont 2001); inverse cubic law (Gopikrishnan et al. 1998); t df 3 to 5.5	t(5) innovations; optional jumps	kurtosis > 1.5 in >= 80 percent of seeds; Hill index 2.5 to 5
Mispricing form	additive dollars (flat N(0,0.5) phi; bull cumsum N(1.5,0.5); crash delta V + N(0,1.5))	P = V + noise; FOMO drift; P = delta V + eps	Summers 1986; Poterba-Summers 1988 (transitory component sd 15 to 25 percent); De Long et al. 1990; Brock-Hommes 1998; Lux-Marchesi 1999; Franke-Westerhoff 2012	FW recursion in calm phases with single-stock re-estimated parameters; scripted drifts in event phases; one GARCH-scaled innovation; no dollar noise, no clips	ACF(1) of x >= 0.98; sd of x in calm 8 to 20 percent; |ACF(1) of r| < 0.15 in calm phases
Crash: fundamental	linear minus 12 dollars over 80	deterministic linear	EPS collapses about minus 25 percent (2001), about	D_V ~ U(10, 30) percent as log-linear	V trough/peak in [0.70, 0.90]

	d; panic RW; total V about minus 42 percent	decline, alpha = minus 12	minus 40 percent (2008-09)	drift in the deterioration phase; flat after	
Crash: panic discount delta	delta in {0.85, 0.92, 0.95}; changes MDD by <= 0.6 pp	varying crisis severities	Bear markets >= 20 percent since 1950: 11 to 14 episodes, mean minus 30 to minus 34 percent, about 338 days; 2008 minus 56.8 percent over 517 d; 2020 minus 34 percent in 23 d; 1987 minus 20.4 percent in one day; 40 percent of Russell 3000 names had a >= 70 percent drawdown (J.P. Morgan); Mishkin and White 2002	delta in {0.55, 0.70, 0.85}: MDD about 1 minus delta (1 minus D_V), i.e. minus 56 / minus 44 / minus 32 percent at D_V = 0.20 (ranges minus 51 to minus 62, minus 37 to minus 51, minus 24 to minus 41 over D_V 0.10 to 0.30); panic length U(15, 70) d, front-loaded; stabilisation toward delta_end ~ U(delta, 1)	regress MDD on delta controlling for D_V: partial R2 > 0.7; MDD spread >= 20 pp between delta 0.55 and 0.85; worst day minus 6 to minus 15 percent; panic daily sigma 3 to 5 percent
Crash: volatility	linspace 1 to 1.5 to 3 to 2, identical across seeds	volatility mean-reverts	realised S&P vol about 80 percent a year Oct 2008, about 90 percent Mar 2020; VIX records 80.86 / 82.69	GARCH omega x 4 in panic (sensitivity 3 to 6), x 1.5 in stabilisation; decays by persistence	20-day realised vol 50 to 100 percent a year in panic; by the end of stabilisation at least 30 percent below its panic peak and below 50 percent annualised
Bubble: run-up	cumsum dollars; floors; +126 percent; P/V 2.2; sigma 1.3 percent/day	regime-switching momentum (Hamilton 1989); exponential disconnect	Greenwood, Shleifer, You 2019: 40 industry run-ups >= 100 percent, 21 crashed; crash probability 20 / 53 / 80 percent after 50 / 100 / 150 percent run-ups; crashes predicted by rising volatility, share issuance and price acceleration (turnover is elevated in run-ups but is not a significant crash predictor in GSY's regressions); Nasdaq +400 then minus 78 percent; Sornette LPPL 0 < m < 1; Sornette et al. 2018: about two thirds of crashes followed a volatility lull	compounding drift g_t (2.1 block 3) with a hazard top h_t = h_0 exp(b x_t) calibrated to P(top in horizon) about 0.5 and peak P/V 1.6 to 2.5 in topped seeds; volume tied to x; omega x 1.5 (mania) to 2 (blow-off); post-top leg minus 30 to minus 50 percent in 10 to 30 d at omega x 3	log-price convex over the mania run (mean second difference > 0); topped seeds 40 to 60 percent; peak P/V 1.6 to 2.5 in topped seeds; corr(volume, |r|) > 0; both strata reported
Bubble: value	plateau	Value plateaus	GSY 2019: fundamentals rise in run-ups	V grows at mu_V throughout	cross-seed mean of V_T / V_0 in [1.0, 1.15]
Sustained bull (new regime, E24)	n/a	n/a	Bull regimes average about 25 months (Pagan and Sossounov 2003); 19 of GSY's 40 run-ups did not crash	V drift U(0.0015, 0.0025)/day over the episode; d_t = 0 so x stays near 0; no top	x within [minus 0.10, +0.15]; final return +20 to +80 percent
Sentiment	i.i.d. N(0, 0.3); phase means	same	daily news sentiment autocorrelated; Tetlock	s_t = m_t + 0.85 (s_t-1 minus m_t-1) +	ACF(1) in [0.7, 0.9]; corr(s_t,

	+0.7 / minus 0.8		2007: one sd of pessimism gives minus 8.1 bp next-day DJIA, about 6.8 bp reversed over days 2 to 5; Brown and Cliff 2004: survey sentiment correlates with contemporaneous returns, little prediction; RavenPack index corr with S&P level about 0.79; AAII weekly AR(1) about 0.7 to 0.8 (approx.); Baker-Wurgler monthly > 0.9 (approx.)	0.25 r_t/sigma_r + 0.25 eps, tanh-squashed, where the offset m_t is a function of the price path, NOT of the phase label: m_t = 0.6 tanh(2 x_t) + 0.3 tanh(20-day return / 0.15) (so mania and panic sentiment emerge from x and returns); plus a configurable predictive component b_pred: default +8 bp expected next-day return per +1 sd of sentiment (pessimism of one sd gives minus 8 bp), with 6 bp reversed over days 2 to 5, per Tetlock; b_pred = 0 as a control	r_t) 0.25 to 0.55; within-phase (or flat-only) partial corr(s_t-1, r_t) equals the configured b_pred within CI; no phase information beyond what x and returns carry (L2b)
Earnings / P/E	EPS = V/15 daily; P/E capped 200	EPS = V/15	quarterly EPS about a seasonal random walk with drift (Foster 1977; Ball and Brown 1968); announcements 25 to 45 d after quarter end; analyst quarterly error several percent of EPS (approx.); S&P trailing P/E mean 19.4 median 17.7 (1971 to 2017); CAPE median about 16 to 17; single-stock P/E IQR roughly 12 to 30 (approx.)	hidden multiple k ~ U(14, 22) per seed; EPS_q = V at quarter end / k x exp(N(0, 0.10)), announced at quarter end + U(25, 35) d; trailing-4Q P/E (warm from the 260-day burn-in); analyst fair-value estimate F_t = V_t exp(u_t), u AR(1) rho 0.95 sd 0.15, weekly, with the error sd FIXED ex ante (not tuned to the audit); P/E cap 200	L1 to L3 of Section 5 per phase; quarterly steps visible
Dividend	0.4 EPS / P daily	40 percent payout	S&P payout about 35 percent; yield 1.6 to 2 percent long-run; dividends sticky (Lintner 1956)	payout 0.35, sticky: DPS_q = 0.7 DPS_q-1 + 0.3 payout x EPS_q; yield = 4 DPS / P	yield 1 to 4 percent except panic; does not move daily with V
Implied volatility	15 x vol_regime; deterministic per phase	forward-looking proxy	VIX 1990 to 2026 mean about 19.4, median about 17.6 (FRED); records 80.86 (20 Nov 2008), 82.69 (16 Mar 2020); variance risk premium +3 to 4 vol points on average, widening in crises (Carr and Wu 2009; Bollerslev, Tauchen, Zhou 2009); single-stock IV 25 to 45 percent	IV = sqrt(252) x 21-day GARCH forecast x (1 + premium), premium 0.20 (0.35 in panic), floor 12 percent	corr(IV, next-20-day realised vol) 0.4 to 0.8; IV minus RV +3 to +8 pts in calm, +10 to +25 pts in panic; varies across seeds
Volume	1e6 (1 + 0.5 U) i.i.d. x phase	base volume x phase	Karpoff 1987; Clark 1973; Lo and Wang: daily log	log Vol_t = mu_v + 0.65 (log Vol_t-1	AC(1) log volume 0.5 to

	multipliers	scalar	turnover AC(1) about 0.7; Cboe exchanges' US equity ADV Mar 2020 +129 percent year on year (about 2.3 x)	minus mu_v) + 0.25 (|r_t|/sigma minus 1) + 1.2 |x_t| + 0.30 eps (volume rises with mispricing and with |r|; no label-driven shifts)	0.8; Spearman corr(volume, |r|) 0.2 to 0.5; peak volume ratio 2 to 5 x in panic and blow-off
Technical indicators	SMA20 and SMA50 labelled SMA60; Cutler RSI; MACD no signal; trend regime plus or minus 2 percent	SMA20, SMA50 scaled to T	standard 20/50/200; Wilder 1978; Appel 12/26/9; Brock, Lakonishok, LeBaron 1992	rename to SMA50; Wilder RSI; MACD signal; keep plus or minus 2 percent; all warm from burn-in	unit tests vs TA-Lib
Phases	fixed 80/60/60 days	tau1 = 0.4 T; Hamilton 1989	Hamilton 1989; Ang and Timmermann 2012; Ang and Bekaert 2002; Pagan and Sossounov 2003	schedule of 2.1 block 6: L1 ~ U(0.25 T, 0.55 T); crash L2 U(15, 70); mania until hazard top or horizon; mirrored event-first episodes; jitter; phase-free flat and T = 800 controls	on the mixed set (setup-first, event-first, phase-free): corr(day, phase id) < 0.9; macro-phase classifier from day alone < 80 percent; L2b selectivity (Section 5)
Initial allocation (A1)	100 percent cash for everyone	not stated	design choice	factor: start at own target-band centre (primary) or common 0.5 (secondary); baselines from the same start; cross-persona return comparisons only on the common start	unit test that the start was applied; the bidirectional effect survives both starts or is reported as start-dependent (A1 x A2 factorial, Section 11.3)
Action semantics (A2)	BUY = cash x pct; SELL = holdings x pct	A_t = {a, q, r}	design choice	target cash share; derived labels with a 1-point dead band; same-day close execution (next-open sensitivity); 5 bp per trade; v1 interface behind a flag	SELL feasible at t = 1; any allocation reachable in one step
Horizon disclosure (C12)	agent sees 'Day-N', never T	not stated	terminal-horizon effects on allocation (lifecycle literature)	declare; factor: T known vs unknown (default unknown, as now)	headline effects reported under both
Observation field order (C13)	fixed	not stated	prompt position sensitivity (Liu et al. 2024)	canonical order by default; an ordering-sensitivity arm with per-seed randomised order	effect sizes stable across orders
Frictions and visibility (C15)	none; same-day price	not stated	US retail commission 0; S&P 500 bid-ask cost about 4.5 bp per trade (Nasdaq 2024); institutional median about 6 bp (Frazzini, Israel, Moskowitz); CLQT low tier 2 + 0.5 + 1 bp	default 5 bp x |traded value|; sensitivity {0, 5, 20} bp; cost shown to the agent vs hidden as a factor (default hidden, as now); fractional shares	turnover and cost per run; pre-registered: the flat-market 'trading without signal' result may shrink or invert under

				declared	visible costs

Cont 2001: doi.org/10.1080/713665670 | Engle 2001: doi.org/10.1257/jep.15.4.157 | Hansen and Lunde 2005: doi.org/10.1002/jae.800 | GJR 1993: doi.org/10.1111/j.1540-6261.1993.tb05128.x | Greenwood, Shleifer, You 2019: doi.org/10.1016/j.jfineco.2018.09.002 | Lux and Marchesi 1999: doi.org/10.1038/17290 | Tetlock 2007: doi.org/10.1111/j.1540-6261.2007.01232.x | Ang and Timmermann 2012: doi.org/10.1146/annurev-financial-110311-101808 | Pagan and Sossounov 2003: doi.org/10.1002/jae.664 | Campbell, Lettau, Malkiel, Xu 2001: doi.org/10.1111/0022-1082.00318 | Frazzini, Israel, Moskowitz: papers.ssrn.com/sol3/papers.cfm?abstract_id=2294498

4. Allocation targets and the primary measurement

4.1 Deriving targets from Jiang, Peng and Yan (JFE 2024): the arithmetic (fact-checked)

AAII survey, N = 3,325, median wealth 3.5 million dollars, 93 percent male, mean age 68; replications in HILDA and GSOEP. Equity share of financial wealth per trait point (Table 7): Neuroticism minus 1.74 pp (minus 2.55 in retirement accounts), Openness +0.94 (+1.50), Conscientiousness minus 1.32, Extraversion minus 0.33 (n.s.), Agreeableness minus 0.46 (n.s.); adjusted R2 0.05 to 0.08. One trait point is about one standard deviation. Mapping the paper's O1 and O2 profiles onto the AAII 10th and 90th percentiles (recomputed independently in the fact-check):

Specification	Conservative (O1)	Balanced	Aggressive (O2)	Spread
AAII total equity share (T7 c1)	minus 4.1 pp	+0.2	+3.3 pp	7.4 pp
AAII retirement accounts (T7 c2)	minus 5.4	+0.1	+5.4	10.8 pp
AAII non-retirement (T7 c3)	minus 3.0	+0.2	+3.0	6.0 pp
AAII with belief and risk-aversion controls (T7 c4)	minus 3.6	+0.2	+2.3	5.9 pp
HILDA one-person households (T8 c1)	minus 1.9	0.0	+1.8	3.7 pp
Ceiling: all five traits at P10/P90 in the favourable direction				about 11.5 pp (total), about 12 pp (retirement)

Conclusion: the JFE relationships imply a conservative-to-aggressive cash spread of about 0.06 to 0.12 around a centre near 0.35 to 0.40 cash. The objection to the current targets is NOT the JFE magnitude (any practitioner-anchored scheme, including the v2 bands with a 0.70 spread, is equally far from JFE by design); it is the 0 percent risky endpoint and the claim of Markowitz grounding (4.2). MBTI has no Neuroticism scale (McCrae and Costa 1989: E-I to Extraversion r = minus 0.74, S-N to Openness 0.72, T-F to Agreeableness 0.44, J-P to Conscientiousness minus 0.49) and Neuroticism is the largest JFE coefficient; all three personas are 'J' (higher Conscientiousness lowers equity in the JFE) and INTJ and ENTJ differ only on Extraversion, which is insignificant; on the LLM side Big Five self-reports do not predict behaviour (Kocielnik 2026) and trait prompts move GPT-4o's risk parameters only inside the risk-neutral regime (Hartley 2025; Openness to gain-risk-taking rho = 0.52). Verdict: JFE supports the ordering and a single-digit spread; it cannot be the source of point targets. Use it as a sensitivity and ordering check.

4.2 Markowitz does not give 100 percent cash

Merton/Samuelson risky share = (mu minus r) / (gamma sigma^2). Over the grid equity premium 4.3 to 6 percent x volatility 16 to 20 percent: gamma = 8 to 10 gives 11 to 29 percent risky; gamma = 3 to 4 gives 27 to 78 percent; gamma = 2 gives 54 to 117 percent. At the Vanguard long-run pair (6 percent, 17.5 percent): gamma 10 gives 20 percent, gamma 3 gives 65 percent, gamma 2 gives 98 percent. No finite gamma yields 0 percent. 'C_ideal = 1.0 grounded in Markowitz (1952)' is incorrect as stated. Sources: Damodaran implied ERP Jan 2025 (4.33 percent); DMS/UBS Yearbook 2025 (4.3 percent over bills); Vanguard cash framework (6 percent over cash, sigma 17.5 percent).

4.3 Practitioner, regulatory and survey anchors (all fact-checked against the primary pages unless marked)

Source	Conservative	Balanced	Aggressive
Morningstar US fund categories (Apr 2025)	15 to 30 percent equity	50 to 70 percent	> 85 percent
Morningstar target-risk indexes	20 percent	60 percent	95 percent
Vanguard LifeStrategy	Income 20/80; Conservative Growth 40/60	Moderate Growth 60/40	Growth 80/20
Fidelity target asset mixes	Conservative 20 percent stock (Short-Term mix = 100 percent cash only for fully funded short goals)	Balanced 50 percent	Aggressive Growth 85; Most Aggressive 100
Betterment / Wealthfront	0 percent stocks only when the goal is fully liquidated at horizon	about 56 percent at retirement age	90 percent for 20+ years; risk score 8 about 82 percent equity
FINRA 2111 / ESMA MiFID II	categories, no numeric bands		
SCF studies	risk category moves stock share about 17 pp per step (Gilliam et al. 2010); 25 to 42 pp between extremes (Fieberg et al. 2025); 'no risk' households still hold risky assets (Schooley and Worden 1996, ratio 0.72 on a broad measure; approx., source not retrievable)		
HNWI cash (Capgemini 2024/25)	25 to 26 percent cash and equivalents		

4.4 Targets: stipulated bands, JFE as sensitivity

Persona	Risk category	Cash band	Point C_ideal (band centre)	Justification
ISFJ / O1	Conservative with market exposure	0.70 to 0.90	0.80	Conservative = 15 to 30 percent equity (Morningstar, Fidelity, Vanguard Income); Merton gamma 8 to 10; the prompt names index funds, bonds and sector leaders; no convention places a conservative investor at 0 percent risky. Keep 1.0 only as an explicitly labelled liquidity condition in the stated-target track.
INTJ	Balanced	0.40 to 0.60	0.50	Fidelity Balanced 50, Vanguard Moderate Growth 60/40, Morningstar Moderate 50 to 70; Merton gamma 3 to 4.
ENTJ / O2	Aggressive	0.00 to 0.20	0.10	Aggressive >= 85 percent equity (Morningstar, Fidelity, Vanguard Growth 80/20, Betterment 90); Merton gamma <= 2. (0.20 is the band edge, not an alternative point.)

Band-MAS = mean over t of max(0, |C_t minus centre| minus half-width), reported alongside point-MAS; JFE sensitivity reported as the fraction of models whose conservative-minus-aggressive gap exceeds the JFE spread (0.06 to 0.12) and the practitioner spread (0.60 to 0.70). Under multi-asset the band applies to the total cash share; weights within the risky sleeve are unconstrained.

4.5 The primary measurement: target-free mandate salience (review D17, promoted)

The strongest measurement in this plan needs no C_ideal and therefore survives every objection to MAS (tautology, degenerate optimum, cross-persona incomparability, the Markowitz and JFE magnitude problems). It is Ross and Lo's input-sensitivity criterion applied over time. It is computed in both prompt tracks (Track A: band stated in the prompt; Track B: persona text only) and is the primary measurement; MAS (point and band) is secondary.

11. Design requirements: all personas and arms face identical market paths (same seeds), so the surrogate can attribute allocation to the prompt; many runs per window.
12. Per model, per window w (25-day windows), fit a pre-registered surrogate for the chosen cash share c*_t: features = persona factor (ENTJ, INTJ, ISFJ, no-persona), directive factor (none, declarative placebo, directive placebo, wrapper-only, mandate ISFJ-text, mandate ENTJ-text, ...), rendered market features, and start allocation; PORTFOLIO STATE IS EXCLUDED from the primary specification because it is a function of the persona's own past actions and would absorb the persona effect as portfolios diverge, manufacturing 'decay'; a secondary specification adds portfolio state as a separate block to measure how much of the persona effect is mediated by accumulated state. Seed is NOT a feature (it is collinear with the market features); it is the grouping variable for blocked cross-validation. Method: random forest or ridge; permutation importance with bootstrap CIs; report the persona share S_persona(w), the directive share S_directive(w), and R2.
13. Validation of the measure: a label-permutation null (shuffle persona labels across runs; S_persona must fall to about 0) and the O3 numerical-only arm (S_directive high). The no-mandate, no-persona trader is the reference level of both factors, not a separate validation.
14. Interpretation: under a stateless agent the decision function is identical at every step, so a change in S_persona across windows can only reflect market features becoming more dispersed (event phases) or, in the secondary specification, portfolio mediation; label it phase and state re-weighting. 'Decay' is reserved for the stateful arm (Section 6 and 11.1), where S_persona falling across windows while the market-feature share rises, on the common-start design, is the operational definition. The swapped-mandate arm is read off directly: does the allocation follow the directive factor or the persona factor?
15. Report alongside: self-anchored drift mean |C_t minus C_0| (common-start design only); band-MAS; and the t = 0 separability gate below. Pre-register the feature set, window length and importance method.

4.6 The t = 0 separability gate: run on v1 data now (review F26)

This is a design input, and the v1 data already answer it: day-1 cash 0.88 / 0.92 / 0.92 for ENTJ / INTJ / ISFJ; ISFJ vs INTJ indistinguishable over days 1 to 5 (Mann-Whitney p = 0.356); only ENTJ separates, weakly; no persona near its target. Under v1 the personas as written fail the gate. Consequences: persona prompts must be strengthened or the target communicated (Track A) if point/band adherence is to mean anything; the target-free measure (4.5) remains interpretable when personas under-separate. For v2 the gate is pre-registered on the COMMON-START design (under start-at-target it passes by construction, because C_0 encodes the persona; there it can only be run on the first-day change delta C_1 = C_1 minus C_0): day-1 allocation ordering cons > bal > aggr (Kruskal-Wallis p < 0.01, pairwise Mann-Whitney, Cliff's delta >= 0.47, band-hit rate >= 80 percent), persona decodable from C_1 alone (AUC >= 0.8), surrogate persona share >= 0.5 with R2 >= 0.5, O3 numerical-only arm as the ceiling. Models failing the gate have no mandate to decay and are excluded or flagged; pass rates per family are a headline persona-realism result.

Jiang, Peng and Yan JFE 2024: doi.org/10.1016/j.jfineco.2023.103776 | McCrae and Costa 1989: doi.org/10.1111/j.1467-6494.1989.tb00759.x | Morningstar categories Apr 2025: advisor.morningstar.com/Enterprise/VTC/MorningstarCategoryClassificationUSFunds_April2025.pdf | Fidelity model portfolios: www.fidelity.com/bin-public/060_www_fidelity_com/documents/fidelity/about-fidelity-model-portfolios.pdf | Betterment methodology: www.betterment.com/resources/asset-allocation-methodology | Vanguard cash framework: corporate.vanguard.com/content/dam/corp/research/pdf/a_framework_for_allocating_to_cash.pdf | Fieberg et al. 2025: www.ifo.de/DocDL/cesifo1_wp11666.pdf | Ross and Lo 2026: arxiv.org/abs/2604.23837 | Kocielnik 2026: arxiv.org/abs/2606.12730 | Hartley et al. 2025: aclanthology.org/2025.findings-acl.1085/

5. Leakage, phase-clock and resolvability audits

A design tension must be stated before the thresholds: the fundamentalist mandate is only playable if mispricing is statistically INFERABLE from what the agent sees, and the 'hidden value' claim is only honest if it is not algebraically RECOVERABLE. The v2 earnings process (hidden multiple, quarterly lag, 10 percent noise) and the analyst estimate (15 percent persistent error, fixed ex ante) are set so that the SIGN of a large mispricing is inferable in event phases while the exact level of V is not. The thresholds below encode that dial per phase; they are pre-registered design choices, and the observables are not tuned to pass them (if the audit fails, the design changes and the change is logged). Audit on the rendered prompt fields, not the internal DataFrame. Precedents: KTD-Fin de-anonymisation probe (top-1 <= 3.0 percent vs 0.3 random, Wilson CIs); Machine Spirits section 3.4; partial-input baselines (Gururangan 2018; Poliak 2018); probe selectivity (Hewitt and Liang 2019). Resolvability threshold theta = 0.05 on |x| (sensitivity 0.03 and 0.08); stated, not derived.

Layer	Test	Acceptance
L1 algebraic	The best algebraic inversion from shown fields (trailing EPS x a multiple prior; DPS / payout prior; the analyst estimate) is compared with V.	No inversion reproduces V: max |V_hat minus V| / V above the generator noise floor on >= 99 percent of steps (currently 0.0000 error).
L2 statistical surrogate (value), per phase	Ridge, gradient-boosted trees and a small MLP from the contemporaneous observation vector (and last 5 observations) to x = log(P/V) and to V; held-out seeds and scenarios; price-and-technicals-only control; shuffled-V control (selectivity).	Calm phases: OOS R2 of x <= 0.30 and sign accuracy <= 0.70 on resolvable steps. Event phases: sign accuracy may be high (intended); require OOS R2 of x < 0.90, MAPE of V_hat >= 10 percent, and selectivity of valuation fields over price-only reported.
L2b composite phase clock (review C11)	The same surrogate family trained to predict the MACRO phase class {calm, down-event, up-event, resolution} from the complete rendered observation vector, compared with the same classifier on price-derived fields only (price, SMAs, RSI, MACD, returns); mixed test set including setup-first, event-first and phase-free runs; jitter included.	Selectivity (full minus price-only accuracy) <= a pre-registered margin, proposed 10 pp and frozen after the E1 calibration run. In v2 sentiment and volume are driven by x and returns rather than the phase label, and IV is a GARCH forecast, so the only remaining label-tied input is the omega multiplier, which the price-only classifier also sees; the test therefore checks that no residual clock exists.
L3 model-side probe	Exact rendered prompt (persona removed) to >= 5 frontier LLMs: estimate V / over- or under-valued and by how much; keyword audit for 'fair P/E of 15' priors; shuffled-V control arm (review F27): the same prompts from runs whose V path is shuffled, giving each LLM's chance baseline.	LLM sign accuracy minus its shuffled-V baseline <= surrogate selectivity + 5 pp; MAPE of V_hat >= 10 percent; 200 probes stratified by scenario x phase x seed, Wilson 95 percent CIs.

L4 resolvability	Fraction of steps with |x| >= theta.	Report per scenario and phase; >= 60 percent in event phases; calm/flat coverage reported honestly (3 percent in v1; expected to rise substantially under persistent mispricing, Section 2.2).
L5 withheld-information oracle (review D18)	A best-achievable-from-observables oracle: a surrogate policy fit on rendered fields only, scored like an agent.	Report the gap between this oracle and the V-oracle as the information the environment deliberately withholds.

Run L2 and L2b as continuous-integration tests whenever the generator or observation space changes; publish the tables.

KTD-Fin: arxiv.org/abs/2605.28359 | Machine Spirits: arxiv.org/abs/2604.18602 | Hewitt and Liang 2019: arxiv.org/abs/1909.03368 | Gururangan et al. 2018: arxiv.org/abs/1803.02324

6. Time versus phase, and what is identifiable with which agent (review A3)

Three environment devices: (1) randomised onset and duration per seed, logged, with analyses in phase-relative and calendar time and a mixed model metric ~ phase + calendar_day + (1 + phase | model) + (1 | seed); (2) mirrored orderings (event-first episodes alongside setup-first); (3) phase-free long horizons (flat at T = 200 and T = 800). Pre-register that temporal claims use windowed statistics, not expanding minima, tested against a constant-exposure null.

What these devices identify depends on the agent. With the current stateless single-turn agent the prompt has constant length and the persona and mandate text sit at the same token offset on day 200 as on day 1; the phase-free T = 800 run is then predetermined to show no elapsed-context effect, because there is no elapsed context. For the stateless agent what remains identifiable is phase dependence and portfolio-state dependence (the ratchet), and the plan says so. An elapsed-context claim needs the agent-side workstream in Section 11.1 (a stateful arm with genuinely accumulating context, and the mandate present in every arm); the environment devices above are necessary for that claim but not sufficient.

7. Harness: experimental arms to build in now (review B4 to B8, B6)

Retrofitting controls costs far more than designing them in. The following are first-class, configurable arms of the harness, not later ablations.

Arm / parameter	What it is	What it settles
Four-arm injection (B4)	static / declarative placebo / directive placebo (length-, delimiter- and imperative-matched, behaviourally irrelevant) / mandate; plus a wrapper-only arm (the 'ACTIVE MEMORY REFRESH' framing with no mandate).	Content vs imperative force vs framing. The current placebo is regulatory boilerplate against a capitalised second-person command with a delimiter.
Swapped mandate (B5)	ISFJ mandate into an ENTJ-personaed agent and vice versa (and into the no-persona trader); start allocation = the persona's target.	Whether effects track injected content or generic instruction pressure. The decisive test; read off the directive vs persona shares in 4.5.
No-mandate, no-persona trader (B6)	System prompt: 'You manage a single-asset portfolio of cash and one stock. Each day choose the target cash share.' plus the I/O contract; one variant with a generic objective ('maximise long-run portfolio value') and one with none; starts at 0.5.	The reference level for both factors in 4.5; tells you how much any persona or directive moves behaviour.

Forked restatement probe (B7)	At intervals, a forked off-transcript call asking the agent to restate its mandate (and, in stateful arms, what it holds and why); the fork never enters the main transcript.	Separates salience decay (cannot restate) from knows-but-violates (restates and violates). Whenever the mandate is in the prompt (memory arm today; every arm after 11.1) a stateless agent will restate it near 100 percent by construction, and that is reported; on today's static arm, which contains no core mandate, the probe measures persona recall only.
Mandate wording as an environment parameter (B8)	Paraphrase, abbreviation, delimiter on/off, imperative strength, removal of action-inciting clauses ('If the trend is up, BUY'); the numeric-only (O3) wording as one level; the ISFJ 'puts/hedges' clause rewritten for the single-asset space.	The tautology risk: whether the ENTJ penalty is the metric reading an instructed action.
Initial allocation (A1), horizon disclosure (C12), field order (C13), cost visibility (C15)	Each a configurable factor with a default and at least one alternative level.	Whether headline effects survive design choices that were previously constants.
Decode replicates (D22)	>= 3 decode replicates per cell at the declared temperature (0.2 in code, not 0.0 as the paper states), with sampling parameters logged.	An estimate of model-output variance, which currently does not exist; seeds reseed only the market.

8. Evaluation layer: baselines, normalisation, scoring rules, costs, reliability

16. Baselines inside the environment for every scenario x seed, all rule-based and free except the no-mandate trader (an LLM): always-HOLD, always-BUY (fully invested), always-SELL (all cash), random (10 seeds), buy-day-1-then-HOLD, constant-mix at each persona band centre with daily rebalancing, momentum (SMA20 > SMA50), mean-reversion (RSI 30/70), V-oracle (target cash share 0 if x < minus theta, 1 if x > +theta, unchanged otherwise), mandate-conditional oracle (the V-oracle clipped to the persona's band), best-achievable-from-observables oracle (Section 5, L5), and the no-mandate trader (Section 7). All baselines start at the same allocation as the cell they are compared with. Precedents: CLQT 8 passive baselines and 'beats k of 8'; Lopez-Lira deterministic agents; OpenPM random-N and always-flat.
17. Normalisation per metric: for RG and return-type metrics, floor = best of {always-HOLD, random, buy-and-hold}, ceiling = mandate-conditional oracle; for MAS the ceiling is the constant-mix policy (MAS = 0 by construction) and the floor is the worst of {always-BUY, always-SELL, random}; note that adherence metrics reward inaction, which is why the zero-trade audit and the target-free measure are reported beside them. Report raw and normalised, and 'beats k of n'.
18. Unresolvable steps (review D19): steps with |x| < theta are not scored right-or-wrong; either excluded with coverage reported, or scored three-valued ('no correct action'); never averaged silently. With 96.9 percent of v1 flat steps unresolvable this changes the flat-market RG entirely.
19. Costs: 5 bp x |traded value| on every trade (re-targeting pays on the delta; HOLD pays nothing); sensitivity {0, 5, 20} bp; cost shown vs hidden as a factor; execution rule as in 2.1; fractional shares declared; daily long-only decisions; log turnover and cost per run.
20. Reliability: parse failures and API fallbacks logged separately and excluded from behavioural metrics (CLQT 7.11); zero-trade shares per cell reported; prompt hash and generator config hash in every output row (F28); the rendered prompt published verbatim in the appendix.
21. v1 to v2 comparison (review D20): the metric axis is a re-scoring of the existing v1 runs (point-MAS with v1 targets; band-MAS with v2 bands; the target-free measure), so metrics x environments is a

full factorial by re-scoring. The start-design and action-interface axes CANNOT be applied to v1 data; they are run on the v2 harness only, and the v2 grid includes a 100-percent-cash-start, v1-interface cell as the bridge to v1.

9. Validation checklist (run on >= 50 seeds per scenario; publish the pass-rate table)

#	Property	Test	Pass criterion
1	No linear autocorrelation of returns in calm phases (Cont 1)	Ljung-Box Q(10) on r within calm phases (or on phase-demeaned returns); ACF(1)	p > 0.05 in >= 80 percent of seeds; |ACF(1)| < 0.15 (scripted event drifts legitimately induce positive ACF and are excluded)
2	Heavy tails (Cont 2, 7)	excess kurtosis; Jarque-Bera; Hill index	kurtosis > 1.5 in >= 80 percent; JB rejects; Hill 2.5 to 5
3	Volatility clustering (Cont 6)	Ljung-Box on |r| and r^2; ARCH-LM(5)	p < 0.01 in >= 80 percent; ACF|r|(1) 0.1 to 0.4
4	Decay of ACF|r| (Cont 8)	descriptive only, on T = 800 paths	reported; not a pass criterion (GARCH decay is exponential and 200-day ACFs beyond lag 20 are noise)
5	GARCH persistence recoverable	refit GARCH(1,1) per path	median alpha + beta in [0.90, 0.995]
6	Leverage effect (Cont 9)	corr(r_t, |r_t+1|); GJR gamma	negative in >= 70 percent; gamma > 0
7	Volume-volatility (Cont 10)	Spearman corr(volume, |r|); AC(1) log volume; Shapiro-Wilk	0.2 to 0.5; 0.5 to 0.8; not rejected
8	Gain/loss asymmetry (Cont 3)	skewness in crash; worst vs best day	skew < 0; worst day larger in >= 70 percent of crash seeds
9	Mispricing persistence (FW regime)	AR(1) on x in calm; sd of x in calm	ACF(1) >= 0.98; half-life >= 60 days (reported, not bounded above); sd 8 to 20 percent
10	Delta matters	regress MDD on delta controlling for D_V	partial R2 > 0.7; spread >= 20 pp between delta 0.55 and 0.85
11	Bubble shape and populations	mean second difference of log P over the mania run; topped share; peak P/V in topped seeds; metrics by stratum	convex; 40 to 60 percent topped; 1.6 to 2.5; both strata reported
12	Sentiment dynamics	ACF(1); corr(s_t, r_t); within-phase partial corr(s_t-1, r_t)	0.7 to 0.9; 0.25 to 0.55; equals configured b_pred within CI
13	IV realism	mean IV by phase; corr(IV, next-20-day RV); IV minus RV by phase	calm 25 to 35 percent, panic 60 to 100; 0.4 to 0.8; +3 to +8 pts calm, +10 to +25 pts panic; non-degenerate across seeds
14	Value leak	L1, L2 (per phase), L3 of Section 5	calm R2 <= 0.30; event R2 < 0.90 and MAPE >= 10 percent; LLM probe within its shuffled-V baseline + margin; no algebraic inversion
15	Phase/time separability	classifier from day only; corr(day, phase), on the mixed test set	< 80 percent; < 0.9
16	Composite phase clock	L2b selectivity on macro classes	<= the frozen margin (proposed 10 pp)
17	Conditioning	rejection rate per scenario; topped share	rejection < 5 percent; joint conditioning published
18	Start design applied	unit test on C_0 per cell; A1 x A2 factorial run	C_0 as configured; the bidirectional effect reported under both starts and both interfaces
19	Action-space reachability	any target allocation reachable in one step; SELL feasible at t = 1	unit tests
20	Magnitudes	MDD per scenario; daily sigma; worst day	crash MDD minus 20 to minus 65 percent; calm sigma 1.4 to 2.2 percent/day; worst day minus 6 to minus 15 percent in panic

Reference values to print beside the synthetic numbers: S&P 500 daily excess kurtosis about 7 to 10; |r| ACF(1) about 0.2; TwinMarket SSE-50 kurtosis 7.26, leverage 0.14, GARCH alpha + beta 0.95; Hashimoto 18 JPX stocks kurtosis 7.85 plus or minus 1.07, |r| ACF(1) 0.19, |r|-volume correlation 0.46.

10. What the environment can honestly claim, against the precedents

Requirement	Hashimoto 2025	Machine Spirits	ABIDES	CLQT	FinPersona now	FinPersona v2
Documented fundamental process	GBM	analytic y/r	exogenous oracle	n/a	GBM flat; undocumented RW in crash	one equation per scenario, parameter table
Decoupling with theoretical provenance	Chiarella-Iori demand in a LOB	Hommes positive feedback	ZI arbitrage	n/a	hand-coded drifts	FW-shaped dynamics (single-stock re-estimated), regime-scripted events, stated as such
Stylized-facts validation vs Cont 2001	Table 3 vs 18 JPX stocks	RD/RDMAX vs humans	Vyetrenko 2020	n/a	none	20-item table with reference ranges
Leakage audit on the rendered prompt	not probed	section 3.4	n/a	PiT gate	exact leak	L1 to L5, per phase, with CIs and shuffled-V controls
Resolvability coverage and scoring rule	n/a	bubble threshold robustness	n/a	n/a	96.9 percent flat unresolvable, averaged silently	theta stated; coverage per phase; unresolvable steps not scored
Baselines inside the env	FCN-only control	RE solutions	ZI background	8 baselines	none in env or paper (the audit computed three)	12 baselines incl. no-mandate trader and observables-oracle
Seed-reproducible, price and V independent of agent actions	per-agent PRNG; price endogenous	deterministic given forecasts	per-agent PRNG; price endogenous	ledger	yes, but clocks	separate streams; no clocks; config hash per row
Time vs phase identifiable	rolling horizons	bubble-relative timing	n/a	n/a	confounded	randomised onset, mirrored order, phase-free control; agent-side stated
Initial conditions and action semantics declared	n/a	n/a	n/a	n/a	100 percent cash; asymmetric BUY/SELL; undeclared	start as a factor; target-share action; declared
Execution and cost semantics	limit orders	n/a	LOB	4 tiers	undeclared zero cost	tier + sensitivity + visibility factor
Observation space = what is rendered	prompt in appendix	prompt in appendix	n/a	tool logs	6 of 16 fields never shown; SMA mislabelled	Table 2 generated from code; prompt published; hash logged
Regime validity and conditioning reported	n/a	bubble definition	n/a	n/a	invariants that never bind	criteria per scenario, rate, joint conditioning,

						strata
Multi-asset	many agents, one asset	one asset	many	real universe	one	N-asset capable; 3-asset extension

Honest novelty statement for the paper. Not novel: a synthetic single-asset market with an exogenous fundamental (ABIDES oracle, Hashimoto/PAMS, Lopez-Lira dividend fundamental); stylized-facts validation (Hashimoto, TwinMarket); leakage probes (KTD-Fin, Machine Spirits); cost tiers and baselines (CLQT). Novel if delivered: (i) a referee-scored, phase-controlled price-value decoupling whose price path is independent of the agent (no price impact) with per-step ground truth and regimes randomised in onset and duration, with dynamics shaped by a single-stock-calibrated fundamentalist-chartist process (order-book simulators provide per-step ground truth but not an agent-independent price path for a single LLM agent); (ii) a published leakage, phase-clock and resolvability audit with per-phase thresholds on the rendered observation; (iii) a target-free mandate-salience measure plus mandate-conditional oracle and trivial-policy normalisation; (iv) optionally, the multi-asset variant with preserved ground truth. Claim it as a measurement instrument, not as market realism.

11. Parallel workstreams the environment rebuild does not cover (review A3, F25)

11.1 Agent-side: the stateful arm (Path B)

Path B, as defined in docs/cycle2_reviews/FinPersona-Bench_Position_and_Revision_Plan_Update_Aug2026, is the rebuild of the agent so that context genuinely accumulates and the titular phenomenon is instantiated. Requirements that must be specified in parallel with E1, because they change what the environment must log: (1) the context design, chosen explicitly and stated: rolling transcript window, summarisation memory, or full transcript up to a token budget, each of which makes 'context accumulates' mean something different (Li et al. 2024; When Attention Closes; Meta Proactive Memory); (2) the core mandate present in the system prompt of every arm at t = 0 (today the static arm never contains it), with the memory arm additionally re-injecting; (3) the current stateless grid retained as the control arm; (4) per-step logging of the full rendered context length and the mandate's token offset so that M1 (attention to the mandate span) has the covariates it needs; (5) the forked restatement probe (Section 7) as the accessibility instrument; (6) the mechanistic programme (M0 to M4) run on open anchors as the revision plan designed it, pre-registered to publish the accessibility-versus-behaviour dissociation. With this workstream, Section 6's phase-free T = 800 run becomes a genuine elapsed-context control; without it, the environment alone cannot test decay.

11.2 Statistics and reporting

Orthogonal to the environment and not fixed by it: model-level inference with mixed-effects models and random slopes by model (effective N is models, not runs; the pooled crash result fails at model level, 12 of 18, p = 0.238, in v1); Benjamini-Hochberg across the metric family; effect sizes and confidence intervals throughout; the re-scoring of v1 (Section 8, item 6); the degeneracy audit (parse failures, refusals, zero-trade shares, static cash baseline) reported beside every contrast; and decode replicates (Section 7). Name it as a track with its own owner or it falls between the two plans.

11.3 Pre-registration: expected casualties and the spine decision (review D21, F29)

Write down, before E7, which v1 findings the diagnosis predicts will not survive, so the result reads as confirmation rather than salvage:

- Delta sensitivity (Appendix J): MDD was minus 47.5 / minus 47.2 / minus 47.0 across the three deltas; under v2 delta will matter and the 'stable across severities' statement is retired.
- Flat-market RG: 96.9 percent of steps unresolvable; under the new scoring rule the flat RG number changes or is not reported.
- Bull-trap RG levels: both reported arms sit inside the trivial-policy envelope (66.9 to 99.7); the paper's 8.8 percent relative gap (7.5 points) becomes a normalised score and may vanish.
- The bidirectional MAS pattern itself (17 of 18 vs 16 of 18): if A1 (everyone starts at 100 percent cash) is the driver, the split shrinks under the start-at-target design; this test must hold the action interface at v1 (A1 x A2 factorial), because the target-share action removes the ratchet at the same time; under the target-free measure it is re-expressed as a change in directive weight rather than a sign reversal.
- Cross-persona return and drawdown comparisons under the start-at-target design measure starting exposure; they are not reported as persona effects.
- The pooled crash effect and the 4.4x are already withdrawn on statistical grounds and will not be re-claimed.
- Under visible transaction costs the flat-market 'trading without signal' failure mode may shrink or invert.

Spine decision (F29): two defensible theses exist, the instrument (this document) and the content-isolated sign reversal (Cycle 1), and, with the agent-side workstream, a third, the decay/dissociation result. They are not the same paper and they optimise the environment differently: the instrument thesis optimises for leak-free ground truth, resolvability, baselines and multi-asset; the content thesis optimises for the harness arms, persona separability and the target-free measure; the decay thesis optimises for time-vs-phase, the stateful agent and the restatement probe. Most environment work is common to all three, but the team must choose which is the contribution and which is the apparatus before E1 locks the design. The plan's recommendation: the instrument is the apparatus in every case; the thesis is decided by the causal arms (content) and by the stateful grid (decay); write the paper so that the instrument stands even if both theses return nulls.

12. Author review of the first draft (A1 to F29): evaluation and where each item landed

Item	Verdict	Where it is now in the plan
A1 100 percent cash start generates the headline	Accepted; verified in code. The strongest new confound in the review cycle.	Diagnosis 12; 2.1 block 8 (with the new-confound acknowledgement); spec row; checklist 18; casualties; A1 x A2 factorial
A2 BUY/SELL asymmetry, SELL no-op on day 1	Accepted; verified in code.	Diagnosis 13; 2.1 block 9 (target cash share, derived labels, execution, cost); spec row; checklist 19
A3 Path B undefined; T = 800 flat control null by construction	Accepted.	Section 6; Section 11.1
B4 four-arm injection	Accepted (plus wrapper-only arm).	Section 7
B5 swapped mandate	Accepted; start allocation defined.	Section 7; 4.5

B6 no-mandate control	Accepted; prompt specified; reference level of the 4.5 factors.	Sections 7, 8
B7 forked restatement probe	Accepted; statement conditioned on the mandate being present in the prompt.	Section 7; 11.1
B8 mandate wording as environment parameter	Accepted.	Section 7
C9 rejection sampling stacks with the probabilistic top	Accepted with a nuance: referee-side rejection on a stated criterion applied uniformly to an exogenous generator is transparent conditioning; the joint conditioning is published and topped/un-topped seeds are reported as two populations. The sustained-bull regime is a COMPLEMENTARY control (no mispricing; caution is wrong), not a substitute for the un-topped bubble population (large persistent mispricing; caution right but unrewarded in-horizon).	2.1 blocks 3 and 7; spec row 'Bubble'; checklist 11 and 17
C10 FW calibrated on index data	Accepted; upgraded to a decision: re-estimate on single stocks; provenance claimed for the functional form only.	Section 2.2; Section 10 row
C11 composite phase classifier	Accepted with a nuance: phase inference from price dynamics is legitimate; the test is selectivity of non-price fields. Sentiment and volume are now driven by x and returns rather than the phase label so that the test is passable.	Section 5 L2b; spec rows; checklist 16
C12 horizon disclosure	Accepted; verified.	Diagnosis 14; spec row; Section 7 factor
C13 field ordering	Accepted as a sensitivity arm rather than per-seed randomisation by default.	Spec row; Section 7
C14 sentiment predictiveness	Accepted; a configurable choice with a Tetlock-sized default (sign corrected in the fact-check: optimism predicts positive next-day return).	Spec row 'Sentiment'; checklist 12
C15 cost visibility may delete a finding	Accepted; pre-registered.	Spec row; Section 7; casualties
C16 ISFJ mandate outside the action space	Accepted; rewrite for single-asset (B8), defensive low-volatility risky asset under option D.	Diagnosis 15; Section 2 option D; Section 7
D17 target-free salience as primary	Accepted; now the primary measurement with a pre-registered surrogate that excludes portfolio state in its primary specification.	Section 4.5
D18 best-achievable-from-observables oracle	Accepted.	Section 5 L5; Section 8
D19 scoring of unresolvable steps	Accepted; theta stated.	Section 5; Section 8 item 3
D20 metric discontinuity	Accepted; corrected to what is feasible: metrics x environments by re-scoring; start and interface axes on v2 only with a bridge cell.	Section 8 item 6
D21 pre-register casualties	Accepted; list written.	Section 11.3
D22 decode replicates	Accepted.	Section 7
E23 multi-asset	Accepted as option D; conventions specified (cash-share band on total; free weights; low-vol risky defensive asset; per-asset resolvability and oracles).	Section 2 option D; 2.1 block 1; 4.4
E24 mandate-conflict scenario	Accepted as a fourth regime ('sustained bull'), complementary to the un-topped bubble stratum.	2.1 blocks 3 and 6; spec row
F25 statistics workstream absent	Accepted.	Section 11.2
F26 separability gate on v1 data first	Accepted; the v1 answer is already known: the personas as written fail it; the v2 gate is pre-registered on the common-start design.	Section 4.6

F27 shuffled-V control on the LLM probe	Accepted.	Section 5 L3
F28 publish prompts verbatim; config hash	Accepted.	2.1 block 10; Section 8 item 5
F29 two spines, pick one	Accepted as a decision the team must make before E1; recommendation given.	Section 11.3

13. Implementation order (no timelines)

Phase	Work	Exit criterion
E0 Freeze, document, re-score	Tag current code; regenerate Table 2 from code; write the I/O contract; log prompt and config hashes; run the t = 0 separability gate and the metric re-scoring on v1 data; write the pre-registration (casualties, thresholds, feature sets, spine decision).	Paper Table 2 and prompt match the code; pre-registration exists; v1 re-scored under all metric definitions.
E1 Core generator	Separate RNG streams; burn-in; N-asset-capable fundamental GBM with t innovations; FW mispricing re-estimated on single stocks (index parameters as sensitivity; AR(1) fallback behind a flag); scripted event drifts and hazard top; GJR-GARCH-t on the x-innovation with phase multipliers; scheduled regimes with randomised onsets, mirrored orderings, jitter, sustained-bull; rejection sampling with per-scenario criteria and joint conditioning logged; initial allocation as a factor; target cash-share action semantics; no dollar noise, no clips.	Checklist items 1 to 6, 8 to 11, 15, 17 to 20 pass on >= 50 seeds per scenario (item 4 descriptive).
E2 Observables	Quarterly lagged noisy EPS with hidden multiple; trailing P/E; sticky dividends; analyst fair value with ex-ante error; sentiment AR(1) driven by x and returns with configurable predictive component; log-AR(1) volume driven by |r| and |x|; IV from GARCH forecast plus premium; SMA50 rename; Wilder RSI; MACD signal; horizon disclosure and field order as factors.	Checklist 7, 12 to 14, 16 pass; L1 to L5 published per phase.
E3 Harness arms	Four-arm injection plus wrapper-only; swapped mandate; no-mandate trader; forked restatement probe; mandate wording parameters; cost visibility; decode replicates; config-driven arm registry.	Every arm runnable from config on one model; arm definitions published.
E4 Evaluation layer	Baselines and per-metric normalisation; unresolvable-step rule; cost tiers; reliability accounting; band-MAS and point-MAS; target-free salience with the pre-registered surrogate and its permutation null; Track A and B; mandate-conditional regret for RG; stratified bull-trap reporting.	Every metric has floor, ceiling, 'beats k of n'; salience measure passes its permutation null and O3 check.
E5 Agent-side (parallel)	Stateful arm design and implementation; mandate in every arm; context-length and mandate-offset logging; restatement probe hooks; M0 anchors.	Stateless and stateful arms run on the same environment with identical logging.
E6 Calibration report	Stylized-facts table with real-data reference ranges; acceptance rates and joint conditioning; leakage and phase-clock tables; FW single-stock calibration; sensitivity of headline metrics to cost tier and visibility, ISFJ target, delta set, k range, start design, action interface, field order, horizon disclosure.	An appendix a reviewer can check line by line.
E7 Re-run	Main grid at N = 1 (stateless first; stateful when E5 is ready), including the 100-percent-cash / v1-interface bridge cell; 3-asset extension scenario; statistics track applied throughout.	v1 vs v2 comparison table across metric definitions (re-scored) and the v2-only start and interface axes; pre-registered casualties confirmed or refuted.

13.1 Decisions the team must make before E1

- Spine: instrument vs content sign-reversal vs decay/dissociation (Section 11.3).

- Mispricing engine: FW re-estimated on single stocks (recommended) vs index parameters with sensitivity vs the AR(1) fallback.
- Schedule durations as design choices with quoted anchors (recommended) vs a fitted chain.
- Initial allocation primary design: start at own target (recommended) vs common 0.5; both run.
- Action semantics: target cash share (recommended) with the v1 interface kept for the bridge cell.
- ISFJ target 0.80 band (recommended) vs 1.0 kept as a labelled liquidity condition in Track A.
- Bubble top: hazard-based, probabilistic (recommended) with topped and un-topped strata reported, plus the sustained-bull regime as the complementary no-mispricing control; vs a guaranteed top.
- Multi-asset: build N-capable now (recommended); decide whether the 3-asset scenario is in the main paper.
- Cost default 5 bp, hidden from the agent by default, with a visible-cost arm.
- Sentiment predictive component default (Tetlock-sized, recommended) vs zero.
- Horizon disclosure default (undisclosed, with a disclosed arm); field order default (canonical, with a randomised arm); execution default (same-day close, with next-open sensitivity).

13.2 Fact-check outcome and numbers still flagged as approximate

The independent fact-check verified against primary sources: JFE Tables 1a, 7, 8, 9 and the recomputed spreads; FW 2012 parameters and J-test; Pruna et al. 2016; Ross and Lo; GSY 40/21 and 20/53/80; Ang and Timmermann Table 1; Ang and Bekaert; Hamilton; Hashimoto Table 3; TwinMarket Table 4; all Morningstar, Vanguard, Fidelity, Betterment and Wealthfront bands; Gilliam; Fieberg; Capgemini; CLQT Table A1; KTD-Fin; Machine Spirits; Kocielnik; the bear-market, 2008, 2020, 1987, J.P. Morgan and Nasdaq facts; VIX records; Nasdaq 4.5 bp; Damodaran; and every code-level claim. Corrections applied in this version: Tetlock minus 8.1 bp (not 5.5); GSY crash predictors are volatility, issuance and acceleration (not turnover); Hartley Openness-alpha rho 0.52 (not 0.63); JFE ceiling about 11.5 pp; Merton ranges restated over the full grid; GJR 1993 no longer cited for a daily gamma range; arXiv 1604.08824 attributed to Pruna, Polukarov and Jennings; the +129 percent volume figure is Cboe exchanges' ADV; VIX mean/median 19.4 / 17.6. Still approximate (verified only via secondary sources or not at all): McCrae and Costa correlations; Pagan and Sossounov 25/15 months; Engle 2001 digits; Hansen and Lunde IBM result; AAII and Baker-Wurgler AR(1) coefficients; single-stock P/E IQR; analyst EPS error range; typical daily GJR gamma; Schooley and Worden 1996 ratios (all copies returned 403); Filbeck et al. 2005 magnitudes; Schwab and BlackRock bands.

14. Source files

docs/env_research/env_research_targets.md; docs/env_research/env_research_timeseries.md; docs/env_research/env_research_design.md; docs/env_research/diag_gen.py; docs/env_research/envplan_factcheck.md (claim-by-claim verification table); docs/env_research/envplan_coherence.md (67 coherence findings, all addressed in this version). Companion documents: docs/cycle2_reviews/FinPersona-Bench_Cycle2_Verification_RedTeam_Report_Aug2026 and FinPersona-Bench_Position_and_Revision_Plan_Update_Aug2026.